

# COI barcoding and length-adjusted genetic distances identify a mislabeled sushi sample

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## 1. Introduction

- 1.1. Accurate species identification is essential in biology, food safety and conservation, but many organisms are difficult to identify from their appearance alone. DNA barcoding addresses this problem by using a short, standardized gene region as a molecular identifier for species (Herbert et al., 2003). In animals, the standard barcode is a segment of the mitochondrial cytochrome c oxidase subunit I (COI) gene. COI is well suited to this role because it is present in many copies per cell, it can be amplified from a wide range of taxa, and its sequence varies enough to distinguish species while remaining nearly constant within a species. An unknown sample can therefore be identified by comparing its COI sequence with reference sequences from known species in public databases such as GenBank (Sayers et al., 2025), using sequence similarity search tools such as BLAST (Altschul et al., 1990). In fish, COI barcoding was shown to discriminate reliably among the species examined (Ward et al., 2005), and it has since become a standard tool for identifying fish and seafood.

## 2. Materials and Methods

### 2.1. Sample Collection and PCR Amplification

- 2.1.1. A small piece of tissue was saved from a sashimi portion ordered as “Maguro” (tuna) at a local sushi restaurant. DNA was extracted from the tissue, and a 481-nucleotide region of the mitochondrial cytochrome c oxidase subunit I (COI) gene was amplified by PCR using the primers described in Section 2.2. The resulting 481-nucleotide query sequence (5' → 3') was used in all downstream analyses. All analyses were scripted in R (`lab1_analysis.R`) unless stated otherwise. ‘

### 2.2. Primer Design and Verification

- 2.2.1. Primers were designed with Primer3Plus v3.3.0 (Untergasser et al., 2007), a web interface to Primer3 (Untergasser et al., 2012). Due to the primers needing to amplify the exact query sequence, the product size range was fixed at 481-481 base pairs. This forced the forward primer to begin at position 1 and the reverse primer to end at position 481. Primer length was constrained to 18-27 nucleotides (optimum 20 nucleotides) and melting temperature ( $T_m$ ) to 50-75 °C (optimum 60 °C), with a maximum  $T_m$  difference of 5 °C between primers. The annealing temperature was 52 °C. All other settings were left at their defaults. Under these settings, Primer3Plus returned no primer pair, because every candidate forward primer beginning at position 1 exceeded the default maximum hairpin  $T_m$

of 47 °C. The maximum hairpin T<sub>m</sub> was then raised to 55 °C. The primers were then verified in R: the forward primer was confirmed to be identical to positions 1-19 of the query sample, and the reverse primer was identical to the reverse complement of positions 460-481 of the query sample (identical() returned “TRUE” for both).

## 2.3. Species Identification by BLAST

2.3.1. On September 23rd, 2026, the query was searched with the web-based blastn (Altschul et al., 1990) against the NCBI core nucleotide database (core\_nt posted September 14th 2026, with 131,070,239 sequences), which is derived from GenBank (Sayers et al., 2025). The search used megablast (Zhang et al., 2000), which is optimized for highly similar sequences. No organism restriction was applied, so that species not listed on the menu could also be detected. Parameters were word size 28, expect threshold 0.05, match/mismatch scores 1/-2, gap costs (existence 0, extension 2.5), low-complexity filter enabled with soft masking during the lookup phase, and a maximum of 100 target sequences. The hit table was downloaded in CSV format, and the taxonomic composition of the hits was examined with the BLAST Taxonomy report.

## 2.4. Reference Sequences for the Menu Species

2.4.1. Each menu item was assigned a representative species: 1) Hiramé, Olive Flounder (*Paralichthys olivaceus*), 2) Sake, Atlantic salmon (*Salmo salar*), which is widely farmed and commonly served as salmon sushi/sashimi in the United States, 3) Hamachi, Japanese amberjack (*Seriola quinqueradiata*), and 4) Maguro, yellowfin tuna (*Thunnus albacares*), which is commonly served as tuna sushi/sashimi in the United States. For each species, the query was searched with blastn restricted to that species using the Organism field on September 23rd, 2026. These searches used discontinuous megablast (Ma et al., 2002), which is designed for more divergent sequences, with parameters: word size 11, expect threshold 0.05, match/mismatch scores 2/-3, gap costs (existence 5, extension 2), low-complexity filter enabled with soft masking during the lookup phase, and a maximum of 100 target sequences and otherwise the same settings as in Section 2.3.

2.4.2. The top hit for each species was retained as its reference (Table 2): MH032483.1 (*P. olivaceus*), MZ407797.1 (*S. salar*), NC\_016868.1 (*S. quinqueradiata*) and PQ812466.1 (*T. albacares*). The top Nile tilapia (*Oreochromis niloticus*) hit from Section 2.3, MN756471.2, was added as a reference for the identified species. All records were downloaded from GenBank in FASTA format. Since NC\_016868.1 is a complete mitochondrial genome (16,537 base pairs), its COX1 gene was extracted

in R at positions 5,508-7,058 (1,551 base pairs), as annotated in the GenBank feature table.

## 2.5. Multiple Sequence Alignment

2.5.1. The query and the five reference sequences were combined in R into a single FASTA file with short sequence labels. The file was aligned with MAFFT v7.526 (Kato and Standley, 2013) using the `--auto` option, which selected the L-INS-i strategy (iterative refinement using local pairwise alignment information).

## 2.6. Pairwise Genetic Diversity and Nucleotide Diversity

2.6.1. Genetic diversity was estimated in R v4.6.1 (R Core Team, 2026). The alignment was imported with `read.dna`, and pairwise distances were calculated with `dist.dna` in the ape package v5.8.1 (Paradis and Schilep, 2019). Distances were uncorrected p-distances, the proportion of compared sites that differ (`model = "raw"`). Positions with a gap in either sequence were excluded separately for each pair (`pairwise.deletion = TRUE`). Raw counts of differing sites (`model = "N"`) were also calculated for comparison with the per-site measure. Since the sequences differed in length (481-1,551 nucleotides), distances were calculated on two alignments: 1) the full 1,551-column alignment, in which each pair was compared over its own overlapping region, and 2) an alignment trimmed to the 481 columns spanned by the query sample, in which every pair was compared over the same sites. The number of sites compared for each pair was counted with a custom R function. Nucleotide diversity ( $\pi$ ) of the trimmed alignment was calculated with `nuc.div` in peags v1.4 (Paradis, 2010).

## 2.7. Code and Data Availability

2.7.1. The R script, input sequences, alignment and output tables are available at [https://github.com/nlinares-GWU2026/pubh6860\\_lab1.git](https://github.com/nlinares-GWU2026/pubh6860_lab1.git).

# 3. Results

## 3.1. Table 1: Primers

### 3.1.1.

| Primer        | Direction                           | Sequence | Length | Tm      | GC%   | Hairpin Tm | Product Size |
|---------------|-------------------------------------|----------|--------|---------|-------|------------|--------------|
| Left Primer 1 | CGGG<br>CAGA<br>ACTA<br>AGCC<br>AGC | 5' → 3'  | 19 bp  | 61.1 °C | 63.2% | 52.7 °C    | 481 bp       |
| Right         | ATAG                                | 3' →     | 22 bp  | 57.8    | 45.5% | 41.8       | 481 bp       |

|             |                                           |    |  |    |  |    |  |
|-------------|-------------------------------------------|----|--|----|--|----|--|
| Primer<br>1 | AAG<br>GAGT<br>AGTA<br>GTAC<br>TGCG<br>GT | 5' |  | °C |  | °C |  |
|-------------|-------------------------------------------|----|--|----|--|----|--|

3.1.2.

3.2. Unrestricted BLAST Results (Top 10 hits - rest are under

3.2.1. “Lab1-Alignment-HitTable” in `/data`):

3.2.2.

|               |            |       |     |   |   |   |     |    |     |   |     |
|---------------|------------|-------|-----|---|---|---|-----|----|-----|---|-----|
| Query_4677543 | MN756471.2 | 100   | 481 | 0 | 0 | 1 | 481 | 1  | 481 | 0 | 889 |
| Query_4677543 | PZ862264.1 | 98.96 | 481 | 5 | 0 | 1 | 481 | 67 | 547 | 0 | 861 |
| Query_4677543 | PZ597065.1 | 98.96 | 481 | 5 | 0 | 1 | 481 | 42 | 522 | 0 | 861 |
| Query_4677543 | PZ056091.1 | 98.96 | 481 | 5 | 0 | 1 | 481 | 76 | 556 | 0 | 861 |
| Query_4677543 | PX210227.1 | 98.96 | 481 | 5 | 0 | 1 | 481 | 67 | 547 | 0 | 861 |
| Query_4677543 | PX096167.1 | 98.96 | 481 | 5 | 0 | 1 | 481 | 85 | 565 | 0 | 861 |
| Query_4677543 | PQ657235.1 | 98.96 | 481 | 5 | 0 | 1 | 481 | 41 | 521 | 0 | 861 |
| Query_4677543 | PP092849.1 | 98.96 | 481 | 5 | 0 | 1 | 481 | 22 | 502 | 0 | 861 |
| Query_4677543 | PP092840.1 | 98.96 | 481 | 5 | 0 | 1 | 481 | 25 | 505 | 0 | 861 |
| Query_4677543 | PP092839.1 | 98.96 | 481 | 5 | 0 | 1 | 481 | 22 | 502 | 0 | 861 |

3.2.2.1.

3.2.2.2. 100 hits, all *Oreochromis*, top hit MN756471.2 at 100% identity over 100% query cover, the remaining at 98.96%

3.3. Table 2: Menu Species Comparison

| Menu name      | Species                       | Accession   | BLAST % | Query Cover | E value | P-distance to the query |
|----------------|-------------------------------|-------------|---------|-------------|---------|-------------------------|
| Hirame         | <i>Paralichthys olivaceus</i> | MH032483.1  | 78.38%  | 100%        | 2e-108  | 0.2204                  |
| Sake           | <i>Salmo salar</i>            | MZ407797.1  | 81.49%  | 98%         | 1e-126  | 0.1933                  |
| Hamachi        | <i>Seriola quinqueradiata</i> | NC_016868.1 | 78.08%  | 100%        | 2e-109  | 0.2204                  |
| Maguro         | <i>Thunnus albacares</i>      | PQ812466.1  | 81.63%  | 100%        | 9e-131  | 0.1850                  |
| - (identified) | <i>Oreochromis niloticus</i>  | MN756471.2  | 100%    | 100%        | 0.0     | 0.0000                  |

3.4. Table 3: Trimmed p-distance Matrix

3.4.1.

|                                                     | Query_<br>sashimi | Nile<br>Tilapia<br>( <i>O.<br/>niloticu<br/>s</i> ) | Hirame<br>( <i>P.<br/>olivace<br/>us</i> ) | (Sake)<br><i>S. salar</i> | (Hamac<br>hi) <i>S.<br/>quinque<br/>radiata</i> | (Magur<br>o) <i>T.<br/>albacar<br/>es</i> |
|-----------------------------------------------------|-------------------|-----------------------------------------------------|--------------------------------------------|---------------------------|-------------------------------------------------|-------------------------------------------|
| Query_<br>sashimi                                   | 0                 | 0                                                   | 0.2204                                     | .1933                     | 0.2204                                          | 0.1850                                    |
| Nile<br>Tilapia<br>( <i>O.<br/>niloticu<br/>s</i> ) | 0                 | 0                                                   | 0.2204                                     | 0.1933                    | 0.2204                                          | 0.1850                                    |
| Hirame<br>( <i>P.<br/>olivace<br/>us</i> )          | 0.2204            | 0.2204                                              | 0                                          | 0.2266                    | 0.1913                                          | 0.2037                                    |
| (Sake)<br><i>S. salar</i>                           | 0.1933            | 0.1933                                              | 0.2266                                     | 0                         | 0.2037                                          | 0.1996                                    |
| (Hamac<br>hi) <i>S.<br/>quinque<br/>radiata</i>     | 0.2204            | 0.2204                                              | 0.1913                                     | 0.2037                    | 0                                               | 0.1892                                    |
| (Magur<br>o) <i>T.<br/>albacar<br/>es</i>           | 0.1850            | 0.1850                                              | 0.2037                                     | 0.1996                    | 0.1892                                          | 0                                         |

### 3.5. Full vs. Trimmed comparison

3.5.1. The six sequences ranged in length from 481 nucleotides (query) to 1,551 nucleotides (Hirame and Hamachi), so in the full 1,551-column alignment, the number of sites compared for each pair ranged from 481 to 1,551 (Table S1). Distances involving the query were identical in the full and trimmed alignments because the query was compared over the same 481 sites in both. Distances between reference sequences differed between the two alignments. In the full alignment, Hirame and Hamachi differed at 285 of 1,551 compared sites (p-distance = 0.1838), whereas the query and Hirame differed at 106 of 481 sites (p-distance = 0.2204). The Hirame-Hamachi pair therefore had the larger raw count of differences but the smaller p-distance. In the trimmed alignment, Hirame and Hamachi differed at 92 of 481 sites (p-distance = 0.1913).

## 4. Discussion

### 4.1. Species Identity and Mislabeling

4.1.1. The COI barcode shows that the sashimi sold as “Maguro” was not tuna but tilapia. The query was identical to the Nile tilapia (*Oreochromis niloticus*) reference across all 481 sites, and all 100 hits from the unrestricted BLAST search belonged to the genus *Oreochromis*. By contrast, every menu species was highly divergent from the query sample. Yellowfin tuna (*T. albacares*) was the least divergent of the four, yet it still differed at 89 of the 481 sites (p-distance = 0.1850). This is close to nine times the around 2% divergence upper limit for COI variation that is commonly used when comparing within species variation (Herbert et al., 2003). It is also greater than the mean COI divergence reported between fish of different families, which is about 15% (Ward et al., 2005). The remaining menu species were even more distant (0.1933-0.2204). None of the four menu species is therefore plausible to be the source of the sample, and the dish ordered as Maguro was mislabeled.

4.1.2. Seafood mislabeling has been widely documented with DNA barcoding (Wong and Hanner, 2008). A global meta-analysis estimated an average mislabeling rate of about 8% with substantial variation among products (Luque and Donlan, 2019). A single sample cannot show where in the supply chain the substitution happened, or whether it was intentional. It does show that short barcode sequences and public reference databases are enough to detect a substitution that a diner could not recognize by eye or taste.

### 4.2. Accounting for Differences in Sequence Length

4.2.1. The sequences in this study ranged from 481 to 1,551 nucleotides, so a raw count of differences would have been misleading. A raw count depends on how many sites are compared, as well as on how different two sequences are from each other, because a longer overlap gives more sites where differences can accumulate. In full alignment, Hiramé and Hamachi differed at 285 sites compared with 106 between the query sample and Hiramé. By raw count, the Hiramé-Hamachi pair would appear nearly three times as divergent, yet its p-distance (0.1838) was lower than that of the query-Hiramé pair (0.2204). The same bias could distort a species identification, like a reference that covered only a part of the query would accumulate fewer differences simply because fewer sites were compared, making it look like a closer match than it is.

4.2.2. I adjusted for length in two ways. First I used per-site measures (p-distance - proportion of nucleotide sites that differ between two individual sequences and  $\pi$  - the average number of nucleotide differences

per site across all possible pairs of sequences in a population), which divide the number of differences by the number of sites compared. Second, I trimmed the alignment to the 481 columns spanned by the query sample, so that every pair of sequences was compared over the same region. Trimming did not change any distance involving the query sample, because the query sample was the shortest sequence and was always compared over its 481 sites. It changed only the distances between the reference sequences, which had previously been compared over regions of 481 to 1,551 sites. For the same reason, we treated the p-distance as the primary comparison rather than BLAST percent identity. BLAST calculates identity over a local alignment that can exclude poorly matching ends.

#### 4.3. Limitations

- 4.3.1. Several limitations should be noted. First, although the *genus* assignment is unambiguous, the *species* call rests mainly on the single reference that matched the query exactly. The other 99 hits, which included *O. aureus* and several unidentified *Oreochromis* records alongside *O. niloticus*, all differed from the query at the same five sites (98.96% identity). This suggests that the sample carries an uncommon haplotype and that closely related tilapia species can share nearly identical COI sequences.
- 4.3.2. Second, mitochondrial DNA is inherited only through the maternal line, so a COI barcode reveals the maternal lineage alone. Hybrid tilapia, such as the *O. aureus* x *O. niloticus* record among the BLAST hit results, cannot be distinguished from pure *O. niloticus* with this marker.
- 4.3.3. Third, each menu name was represented by a single species and single reference sequence. Other species sold under the same names (for example, blufind rather than yellowfin tuna as Maguro) were not tested, and GenBank records occasionally carry incorrect species labels. It should be noted that three of the four menu references came from voucher specimens (linked to a specific DNA submission to verify an organism's identity).
- 4.3.4. Fourth, uncorrected p-distances underestimate the true divergence between distantly related sequences because they cannot account for multiple substitutions at the same site. A model with corrected distance would produce larger values, but it would not change which species is closest to the query.
- 4.3.5. Finally, the primers were designed to amplify only this exact sequence. The forward primer has a predicted hairpin ( $T_m = 52.7^\circ\text{C}$ ) just above the  $52^\circ\text{C}$  annealing temperature, and the two primers differ in  $T_m$  by  $3.3^\circ\text{C}$ ,

both of which could reduce PCR efficiency. In practice, a seafood survey would use established universal fish COI primers (Ward et al., 2005).

## 5. Conclusion

- 5.1. DNA barcoding of the COI gene identified the sashimi sold as “Maguro” as Nile tilapia (*O. niloticus*), a fish not listed on the menu. The query matched a Nile tilapia reference exactly, while all four menu species differed from it at 18.5% - 22.0% of sites, far beyond the variation expected within a species. The reference sequences differed in length, so genetic distances were calculated as per-site p-distances over a common 481-site region, which avoids the bias that raw counts of differences would introduce. These results show that a short barcode sequence, analyzed with public databases and open-source tools, can reliably detect seafood mislabeling.

## References

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